# 02 Pandas

## Plan for this lecture

1. Pandas Series
2. Pandas DataFrame
3. Pandas functions
4. Exercises

## Introduction to Pandas

(https://upload.wikimedia.org/wikipedia/commons/thumb/e/ed/Pandas_logo.svg/320px-Pandas_logo.svg.png)

- Pandas is a package that supplies data analysis tools in Python.
- DataFrames provide convenient formatting to visualise the data source. Often CSVs, and even NumPy arrays are used as the source for these DataFrames.
- It provides useful functions from summaries and descriptive statistics of data, in addition to SQL joins (GroupBy)
- More documentation available at: https://pandas.pydata.org

## Formatting CSV data in Pandas DataFrames

`pip install pandas`

`python3 -m pip install -U pandas --user`

In [2…
```python
import pandas as pd
pd
```
Out[2…
```
<module 'pandas' from '/Users/nick/Library/Python/3.9/lib/python/site-packages/pandas/__init__.py'>
```

In [3…
```python
import numpy as np
np
```
Out[3…
```
<module 'numpy' from '/Users/nick/Library/Python/3.9/lib/python/site-packages/numpy/__init__.py'>
```

## Pandas Series

- Think of a Pandas Series as a single column of data, or a 1D array if you like.
- The row details can have string labels or integer indices

In [1…
```python
my_list = [10,20,30]
pd.Series(my_list)
```
Out[1…
```
0    10
1    20
2    30
dtype: int64
```

In [8…
```python
labels = ['a','b','c']
my_list = [10,20,30]
pd.Series(my_list,labels)
```
Out[8…
```
a    10
b    20
c    30
dtype: int64
```

Numpy arrays can be converted to a Pandas Series

In [1…
```python
arr = np.array([10,20,30])
pd.Series(arr)
```
Out[1…
```
0    10
1    20
2    30
dtype: int64
```

Python Dictionaries can also be converted to a Pandas Series

In [1…
```python
d = {'a' : 10, 'b' : 20, 'c' : 30}
d
```
Out[1…
```
{'a': 10, 'b': 20, 'c': 30}
```

In [1…
```python
pd.Series(d)
```
Out[1…
```
a    10
b    20
c    30
dtype: int64
```

## Indexing

In [1…
```python
ser1 = pd.Series([1,2,3], index = ['UK', 'USA', 'EU'])
ser1
```
Out[1…
```
UK     1
USA    2
EU     3
dtype: int64
```

In [1…
```python
ser1['UK']
```
Out[1…
```
1
```

In [1…
```python
ser1['USA']
```
Out[1…
```
2
```

## Adding / Concatenation

- Just be aware that indices have to match exactly!
- See below...

In [1…
```python
ser2 = pd.Series([5,3,7], index = ['UK', 'USA', 'ASIA'])
ser2
```
Out[1…
```
UK      5
USA     3
ASIA    7
dtype: int64
```

In [1…
```python
ser1 + ser2
```
Out[1…
```
ASIA    NaN
EU      NaN
UK      6.0
USA     5.0
dtype: float64
```

In [3…
```python
np.add(ser1, ser2)
```
Out[3…
```
ASIA    NaN
EU      NaN
UK      6.0
USA     5.0
dtype: float64
```

In [3…
```python
pd.concat([ser1, ser2])
```
Out[3…
```
UK      1
USA     2
EU      3
UK      5
USA     3
ASIA    7
dtype: int64
```

In [1…
```python
ser3 = pd.concat([ser1, ser2])
ser3
```
Out[1…
```
UK      1
USA     2
EU      3
UK      5
USA     3
ASIA    7
dtype: int64
```

In [1…
```python
pd.value_counts(ser3)
```
```
/var/folders/ry/3hkntqmd6lx9rvtg9q4zp4vr0000gn/T/ipykernel_36540/4098305703.py:1: FutureWarning: pandas.value_counts is deprecated and will be removed in a future version. Use pd.Series(obj).value_counts() instead.
  pd.value_counts(ser3)
```
Out[1…
```
3    2
1    1
2    1
5    1
7    1
Name: count, dtype: int64
```

## Using `datetime` for `pd.Series` indices

- We've seen the use of integer indices and str indices.
- We can also have `datetime` indices!
- `datetime` indices will be useful for Time Series Analysis (which we'll see later in the course)

In [8…
```python
data = pd.Series([10, 20, 30])
dates = pd.to_datetime(['2024-01-01', '2025-04-29', '2023-12-15'])

data.index = dates

print(data)
```
```
2024-01-01    10
2025-04-29    20
2023-12-15    30
dtype: int64
```

In [1…
```python
years = pd.Series([2024, 2025, 2023])
months = pd.Series([1, 4, 12])
days = pd.Series([1, 29, 15])

dates = pd.to_datetime({'year': years, 'month': months, 'day': days})
dates
```
Out[1…
```
0   2024-01-01
1   2025-04-29
2   2023-12-15
dtype: datetime64[ns]
```

We will come back to the use of `datetime` for time series analysis later in the course.

## Pandas DataFrame

In [1…
```python
df = pd.DataFrame({'column 1': [1,2,3,4], 'column 2': [5,6,7,8]})  # Using the pandas package to create a DataFrame.
df
```
Out[1…

| | column 1 | column 2 |
|---|---|---|
| **0** | 1 | 5 |
| **1** | 2 | 6 |
| **2** | 3 | 7 |
| **3** | 4 | 8 |

## Col access

In [1…
```python
df = pd.DataFrame(np.random.randn(3,3),
                  index='A B C'.split(),
                  columns="X Y Z".split())
df
```
Out[1…

| | X | Y | Z |
|---|---|---|---|
| **A** | 1.356425 | -1.304502 | 1.148660 |
| **B** | 0.513187 | -0.737688 | 1.009436 |
| **C** | 0.374554 | 1.565119 | -1.970784 |

In [3…
```python
df['Z']
```
Out[3…
```
A   -1.138860
B    0.253194
C   -0.710403
Name: Z, dtype: float64
```

In [3…
```python
df[['X','Z']]
```
Out[3…

| | X | Z |
|---|---|---|
| **A** | -1.044862 | -1.138860 |
| **B** | 0.297150 | 0.253194 |
| **C** | -0.357124 | -0.710403 |

Add a new column via the subscript operator

In [1…
```python
df['W'] = [0.1, 0.2, 0.3]
df
```
Out[1…

| | X | Y | Z | W |
|---|---|---|---|---|
| **A** | 1.356425 | -1.304502 | 1.148660 | 0.1 |
| **B** | 0.513187 | -0.737688 | 1.009436 | 0.2 |
| **C** | 0.374554 | 1.565119 | -1.970784 | 0.3 |

- Remember that `drop()` returns a view - doesn't affect the original df
- Therefore need to assign this returned view to a df

In [1…
```python
df.drop(['W'], axis = 1)
```
Out[1…

| | X | Y | Z |
|---|---|---|---|
| **A** | 1.356425 | -1.304502 | 1.148660 |
| **B** | 0.513187 | -0.737688 | 1.009436 |
| **C** | 0.374554 | 1.565119 | -1.970784 |

In [1…
```python
df
```
Out[1…

| | X | Y | Z | W |
|---|---|---|---|---|
| **A** | 1.356425 | -1.304502 | 1.148660 | 0.1 |
| **B** | 0.513187 | -0.737688 | 1.009436 | 0.2 |
| **C** | 0.374554 | 1.565119 | -1.970784 | 0.3 |

In [1…
```python
df = df.drop(['W'], axis = 1)
df
```
Out[1…

| | X | Y | Z |
|---|---|---|---|
| **A** | 1.356425 | -1.304502 | 1.148660 |
| **B** | 0.513187 | -0.737688 | 1.009436 |
| **C** | 0.374554 | 1.565119 | -1.970784 |

## Row access

In [3…
```python
df
```
Out[3…

| | X | Y | Z |
|---|---|---|---|
| **A** | -1.044862 | 1.633030 | -1.138860 |
| **B** | 0.297150 | 0.429540 | 0.253194 |
| **C** | -0.357124 | -1.112405 | -0.710403 |

In [1…
```python
df.loc['A']
```
Out[1…
```
X    1.356425
Y   -1.304502
Z    1.148660
Name: A, dtype: float64
```

In [1…
```python
df.iloc[0]
```
Out[1…
```
X    1.356425
Y   -1.304502
Z    1.148660
Name: A, dtype: float64
```

## Basic operations / useful functions

In [1…
```python
df.head(2)
```
Out[1…

| | X | Y | Z |
|---|---|---|---|
| **A** | 1.356425 | -1.304502 | 1.148660 |
| **B** | 0.513187 | -0.737688 | 1.009436 |

In [1…
```python
df.tail(2)
```
Out[1…

| | X | Y | Z |
|---|---|---|---|
| **B** | 0.513187 | -0.737688 | 1.009436 |
| **C** | 0.374554 | 1.565119 | -1.970784 |

In [4…
```python
df.describe()
```
Out[4…

| | X | Y | Z |
|---|---|---|---|
| **count** | 3.000000 | 3.000000 | 3.000000 |
| **mean** | -0.368278 | 0.316722 | -0.532023 |
| **std** | 0.671075 | 1.376190 | 0.712964 |
| **min** | -1.044862 | -1.112405 | -1.138860 |
| **25%** | -0.700993 | -0.341432 | -0.924632 |
| **50%** | -0.357124 | 0.429540 | -0.710403 |
| **75%** | -0.029987 | 1.031285 | -0.228605 |
| **max** | 0.297150 | 1.633030 | 0.253194 |

In [4…
```python
df.info()
```
```
<class 'pandas.core.frame.DataFrame'>
Index: 3 entries, A to C
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype  
---  ------  --------------  -----  
 0   X       3 non-null      float64
 1   Y       3 non-null      float64
 2   Z       3 non-null      float64
dtypes: float64(3)
memory usage: 204.0+ bytes
```

In [ …
```python
df.isna().sum()
```
Out[ …
```
X    0
Y    0
Z    0
dtype: int64
```

In [1…
```python
print(df['X'].count())
```
```
3
```

In [4…
```python
print(df['X'].min())
```
```
-1.0448616047446862
```

In [4…
```python
print(df['X'].max())
```
```
0.29715013870668694
```

In [5…
```python
print(df['X'].mean())
```
```
-0.3682784158646874
```

## Reading / Writing CSV files in/out as DataFrames

- `df = pd.read_csv('your_filepath_here.csv')`

In [1…
```python
ecom_df = pd.read_csv("Ecommerce Purchases.csv")
type(ecom_df)
```
Out[1…
```
pandas.core.frame.DataFrame
```

In [1…
```python
ecom_df.info()
```
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   Address           10000 non-null  object 
 1   Lot               10000 non-null  object 
 2   AM or PM          10000 non-null  object 
 3   Browser Info      10000 non-null  object 
 4   Company           10000 non-null  object 
 5   Credit Card       10000 non-null  int64  
 6   CC Exp Date       10000 non-null  object 
 7   CC Security Code  10000 non-null  int64  
 8   CC Provider       10000 non-null  object 
 9   Email             10000 non-null  object 
 10  Job               10000 non-null  object 
 11  IP Address        10000 non-null  object 
 12  Language          10000 non-null  object 
 13  Purchase Price    10000 non-null  float64
dtypes: float64(1), int64(2), object(11)
memory usage: 1.1+ MB
```

In [1…
```python
ecom_df.describe()
```
Out[1…

| | Credit Card | CC Security Code | Purchase Price |
|---|---|---|---|
| **count** | 1.000000e+04 | 10000.000000 | 10000.000000 |
| **mean** | 2.341374e+15 | 907.217800 | 50.347302 |
| **std** | 2.256103e+15 | 1589.693035 | 29.015836 |
| **min** | 6.040186e+10 | 0.000000 | 0.000000 |
| **25%** | 3.056322e+13 | 280.000000 | 25.150000 |
| **50%** | 8.699942e+14 | 548.000000 | 50.505000 |
| **75%** | 4.492298e+15 | 816.000000 | 75.770000 |
| **max** | 6.012000e+15 | 9993.000000 | 99.990000 |

In [1…
```python
ecom_df [ ecom_df['Purchase Price'] == 0.00 ]
```
Out[1…

| | Address | Lot | AM or PM | Browser Info | Company | Credit Card | CC Exp Date | CC Security Code | CC Provider | Email | Job | IP Address | Langua |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **2876** | 332 Jones Parkways\nEast Katherineville, GA 64230 | 39 GT | AM | Mozilla/5.0 (Macintosh; U; PPC Mac OS X 10_6_9... | Lyons, Diaz and Clark | 4204500444841766 | 01/18 | 38 | VISA 16 digit | jennifer11@baker.com | Biomedical scientist | 223.47.177.174 | |
| **5487** | 465 Mallory Ways\nNorth Rebecca, RI 82734-1160 | 93 OH | PM | Mozilla/5.0 (X11; Linux i686; rv:1.9.6.20) Gec... | Flynn and Sons | 30469912089738 | 09/23 | 236 | Discover | mjohnson@austin.org | Stage manager | 43.99.56.59 | |

In [1…
```python
ecom_df.head()
```
Out[1…

| | Address | Lot | AM or PM | Browser Info | Company | Credit Card | CC Exp Date | CC Security Code | CC Provider | Email | Job | IP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 16629 Pace Camp Apt. 448\nAlexisborough, NE 77... | 46 in | PM | Opera/9.56.(X11; Linux x86_64; sl-SI) Presto/2... | Martinez-Herman | 6011929061123406 | 02/20 | 900 | JCB 16 digit | pdunlap@yahoo.com | Scientist, product/process development | 149.14 |
| **1** | 9374 Jasmine Spurs Suite 508\nSouth John, TN 8... | 28 rn | PM | Opera/8.93.(Windows 98; Win 9x 4.90; en-US) Pr... | Fletcher, Richards and Whitaker | 3337758169645356 | 11/18 | 561 | Mastercard | anthony41@reed.com | Drilling engineer | 15. |
| **2** | Unit 0065 Box 5052\nDPO AP 27450 | 94 vE | PM | Mozilla/5.0 (compatible; MSIE 9.0; Windows NT ... | Simpson, Williams and Pham | 675957666125 | 08/19 | 699 | JCB 16 digit | amymiller@morales-harrison.com | Customer service manager | 132.2 |
| **3** | 7780 Julia Fords\nNew Stacy, WA 45798 | 36 vm | PM | Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_0 ... | Williams, Marshall and Buchanan | 6011578504430710 | 02/24 | 384 | Discover | brent16@olson-robinson.info | Drilling engineer | 30. |
| **4** | 23012 Munoz Drive Suite 337\nNew Cynthia, TX 5... | 20 IE | AM | Opera/9.58.(X11; Linux x86_64; it-IT) Presto/2... | Brown, Watson and Andrews | 6011456623207998 | 10/25 | 678 | Diners Club / Carte Blanche | christopherwright@gmail.com | Fine artist | 24. |

In [1…
```python
ecom_df.sample(3)
```
Out[1…

| | Address | Lot | AM or PM | Browser Info | Company | Credit Card | CC Exp Date | CC Security Code | CC Provider | Email | Job | IP Add |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **9490** | 279 Jones Crescent\nGonzalesstad, MS 13178 | 37 Nj | AM | Mozilla/5.0 (iPod; U; CPU iPhone OS 4_3 like M... | Hernandez-Valencia | 30502181354175 | 01/26 | 228 | Maestro | ureynolds@maxwell.com | Lecturer, further education | 113.70.1 |
| **4093** | USCGC Johnson\nFPO AE 61048 | 22 kl | PM | Opera/8.53.(Windows NT 5.2; en-US) Presto/2.9.... | Nguyen-Flowers | 180067246299847 | 08/24 | 589 | VISA 16 digit | kelseygarcia@wolfe.com | Equality and diversity officer | 40.72.1 |
| **1960** | 83317 Kevin Rue Suite 248\nKarentown, PA 89548... | 29 KF | PM | Mozilla/5.0 (Windows NT 5.01; it-IT; rv:1.9.1.... | Smith PLC | 4365212541663011 | 12/17 | 319 | American Express | mariah78@hotmail.com | Diplomatic Services operational officer | 189.25.3 |

export to csv!

In [1…
```python
ecom_df.to_csv("export.csv")
```

In [7…
```python
url = 'https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data'

iris_df = pd.read_csv(url,
                      delimiter=',',
                      names= ['sepal length', 'sepal width', 'petal length' , 'petal width', 'species'],
                      encoding='utf-8', skiprows=1
           )
iris_df
```
Out[7…

| | sepal length | sepal width | petal length | petal width | species |
|---|---|---|---|---|---|
| **0** | 4.9 | 3.0 | 1.4 | 0.2 | Iris-setosa |
| **1** | 4.7 | 3.2 | 1.3 | 0.2 | Iris-setosa |
| **2** | 4.6 | 3.1 | 1.5 | 0.2 | Iris-setosa |
| **3** | 5.0 | 3.6 | 1.4 | 0.2 | Iris-setosa |
| **4** | 5.4 | 3.9 | 1.7 | 0.4 | Iris-setosa |
| **...** | ... | ... | ... | ... | ... |
| **144** | 6.7 | 3.0 | 5.2 | 2.3 | Iris-virginica |
| **145** | 6.3 | 2.5 | 5.0 | 1.9 | Iris-virginica |
| **146** | 6.5 | 3.0 | 5.2 | 2.0 | Iris-virginica |
| **147** | 6.2 | 3.4 | 5.4 | 2.3 | Iris-virginica |
| **148** | 5.9 | 3.0 | 5.1 | 1.8 | Iris-virginica |

149 rows × 5 columns

## DataFrames for CSVs / JSON

- So let's see if we can convert our iris `numpy.array` to a pandas DataFrame to give us a view of the data.

In [7…
```python
url = 'https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data'
iris = np.genfromtxt(url,
                     delimiter=',',
                     names=['sepal length', 'sepal width', 'petal length' , 'petal width', 'species'],
                     dtype=[np. float64, np.float64, np.float64, np.float64, 'U15']
        )
iris
```
Out[7…
```
array([(5.1, 3.5, 1.4, 0.2, 'Iris-setosa'),
       (4.9, 3. , 1.4, 0.2, 'Iris-setosa'),
       (4.7, 3.2, 1.3, 0.2, 'Iris-setosa'),
       (4.6, 3.1, 1.5, 0.2, 'Iris-setosa'),
       (5. , 3.6, 1.4, 0.2, 'Iris-setosa'),
       (5.4, 3.9, 1.7, 0.4, 'Iris-setosa'),
       (4.6, 3.4, 1.4, 0.3, 'Iris-setosa'),
       (5. , 3.4, 1.5, 0.2, 'Iris-setosa'),
       (4.4, 2.9, 1.4, 0.2, 'Iris-setosa'),
       (4.9, 3.1, 1.5, 0.1, 'Iris-setosa'),
       (5.4, 3.7, 1.5, 0.2, 'Iris-setosa'),
       (4.8, 3.4, 1.6, 0.2, 'Iris-setosa'),
       (4.8, 3. , 1.4, 0.1, 'Iris-setosa'),
       (4.3, 3. , 1.1, 0.1, 'Iris-setosa'),
       (5.8, 4. , 1.2, 0.2, 'Iris-setosa'),
       (5.7, 4.4, 1.5, 0.4, 'Iris-setosa'),
       (5.4, 3.9, 1.3, 0.4, 'Iris-setosa'),
       (5.1, 3.5, 1.4, 0.3, 'Iris-setosa'),
       (5.7, 3.8, 1.7, 0.3, 'Iris-setosa'),
       (5.1, 3.8, 1.5, 0.3, 'Iris-setosa'),
       (5.4, 3.4, 1.7, 0.2, 'Iris-setosa'),
       (5.1, 3.7, 1.5, 0.4, 'Iris-setosa'),
       (4.6, 3.6, 1. , 0.2, 'Iris-setosa'),
       (5.1, 3.3, 1.7, 0.5, 'Iris-setosa'),
       (4.8, 3.4, 1.9, 0.2, 'Iris-setosa'),
       (5. , 3. , 1.6, 0.2, 'Iris-setosa'),
       (5. , 3.4, 1.6, 0.4, 'Iris-setosa'),
       (5.2, 3.5, 1.5, 0.2, 'Iris-setosa'),
       (5.2, 3.4, 1.4, 0.2, 'Iris-setosa'),
       (4.7, 3.2, 1.6, 0.2, 'Iris-setosa'),
       (4.8, 3.1, 1.6, 0.2, 'Iris-setosa'),
       (5.4, 3.4, 1.5, 0.4, 'Iris-setosa'),
       (5.2, 4.1, 1.5, 0.1, 'Iris-setosa'),
       (5.5, 4.2, 1.4, 0.2, 'Iris-setosa'),
       (4.9, 3.1, 1.5, 0.1, 'Iris-setosa'),
       (5. , 3.2, 1.2, 0.2, 'Iris-setosa'),
       (5.5, 3.5, 1.3, 0.2, 'Iris-setosa'),
       (4.9, 3.1, 1.5, 0.1, 'Iris-setosa'),
       (4.4, 3. , 1.3, 0.2, 'Iris-setosa'),
       (5.1, 3.4, 1.5, 0.2, 'Iris-setosa'),
       (5. , 3.5, 1.3, 0.3, 'Iris-setosa'),
       (4.5, 2.3, 1.3, 0.3, 'Iris-setosa'),
       (4.4, 3.2, 1.3, 0.2, 'Iris-setosa'),
       (5. , 3.5, 1.6, 0.6, 'Iris-setosa'),
       (5.1, 3.8, 1.9, 0.4, 'Iris-setosa'),
       (4.8, 3. , 1.4, 0.3, 'Iris-setosa'),
       (5.1, 3.8, 1.6, 0.2, 'Iris-setosa'),
       (4.6, 3.2, 1.4, 0.2, 'Iris-setosa'),
       (5.3, 3.7, 1.5, 0.2, 'Iris-setosa'),
       (5. , 3.3, 1.4, 0.2, 'Iris-setosa'),
       (7. , 3.2, 4.7, 1.4, 'Iris-versicolor'),
       (6.4, 3.2, 4.5, 1.5, 'Iris-versicolor'),
       (6.9, 3.1, 4.9, 1.5, 'Iris-versicolor'),
       (5.5, 2.3, 4. , 1.3, 'Iris-versicolor'),
       (6.5, 2.8, 4.6, 1.5, 'Iris-versicolor'),
       (5.7, 2.8, 4.5, 1.3, 'Iris-versicolor'),
       (6.3, 3.3, 4.7, 1.6, 'Iris-versicolor'),
       (4.9, 2.4, 3.3, 1. , 'Iris-versicolor'),
       (6.6, 2.9, 4.6, 1.3, 'Iris-versicolor'),
       (5.2, 2.7, 3.9, 1.4, 'Iris-versicolor'),
       (5. , 2. , 3.5, 1. , 'Iris-versicolor'),
       (5.9, 3. , 4.2, 1.5, 'Iris-versicolor'),
       (6. , 2.2, 4. , 1. , 'Iris-versicolor'),
       (6.1, 2.9, 4.7, 1.4, 'Iris-versicolor'),
       (5.6, 2.9, 3.6, 1.3, 'Iris-versicolor'),
       (6.7, 3.1, 4.4, 1.4, 'Iris-versicolor'),
       (5.6, 3. , 4.5, 1.5, 'Iris-versicolor'),
       (5.8, 2.7, 4.1, 1. , 'Iris-versicolor'),
       (6.2, 2.2, 4.5, 1.5, 'Iris-versicolor'),
       (5.6, 2.5, 3.9, 1.1, 'Iris-versicolor'),
       (5.9, 3.2, 4.8, 1.8, 'Iris-versicolor'),
       (6.1, 2.8, 4. , 1.3, 'Iris-versicolor'),
       (6.3, 2.5, 4.9, 1.5, 'Iris-versicolor'),
       (6.1, 2.8, 4.7, 1.2, 'Iris-versicolor'),
       (6.4, 2.9, 4.3, 1.3, 'Iris-versicolor'),
       (6.6, 3. , 4.4, 1.4, 'Iris-versicolor'),
       (6.8, 2.8, 4.8, 1.4, 'Iris-versicolor'),
       (6.7, 3. , 5. , 1.7, 'Iris-versicolor'),
       (6. , 2.9, 4.5, 1.5, 'Iris-versicolor'),
       (5.7, 2.6, 3.5, 1. , 'Iris-versicolor'),
       (5.5, 2.4, 3.8, 1.1, 'Iris-versicolor'),
       (5.5, 2.4, 3.7, 1. , 'Iris-versicolor'),
       (5.8, 2.7, 3.9, 1.2, 'Iris-versicolor'),
       (6. , 2.7, 5.1, 1.6, 'Iris-versicolor'),
       (5.4, 3. , 4.5, 1.5, 'Iris-versicolor'),
       (6. , 3.4, 4.5, 1.6, 'Iris-versicolor'),
       (6.7, 3.1, 4.7, 1.5, 'Iris-versicolor'),
       (6.3, 2.3, 4.4, 1.3, 'Iris-versicolor'),
       (5.6, 3. , 4.1, 1.3, 'Iris-versicolor'),
       (5.5, 2.5, 4. , 1.3, 'Iris-versicolor'),
       (5.5, 2.6, 4.4, 1.2, 'Iris-versicolor'),
       (6.1, 3. , 4.6, 1.4, 'Iris-versicolor'),
       (5.8, 2.6, 4. , 1.2, 'Iris-versicolor'),
       (5. , 2.3, 3.3, 1. , 'Iris-versicolor'),
       (5.6, 2.7, 4.2, 1.3, 'Iris-versicolor'),
       (5.7, 3. , 4.2, 1.2, 'Iris-versicolor'),
       (5.7, 2.9, 4.2, 1.3, 'Iris-versicolor'),
       (6.2, 2.9, 4.3, 1.3, 'Iris-versicolor'),
       (5.1, 2.5, 3. , 1.1, 'Iris-versicolor'),
       (5.7, 2.8, 4.1, 1.3, 'Iris-versicolor'),
       (6.3, 3.3, 6. , 2.5, 'Iris-virginica'),
       (5.8, 2.7, 5.1, 1.9, 'Iris-virginica'),
       (7.1, 3. , 5.9, 2.1, 'Iris-virginica'),
       (6.3, 2.9, 5.6, 1.8, 'Iris-virginica'),
       (6.5, 3. , 5.8, 2.2, 'Iris-virginica'),
       (7.6, 3. , 6.6, 2.1, 'Iris-virginica'),
       (4.9, 2.5, 4.5, 1.7, 'Iris-virginica'),
       (7.3, 2.9, 6.3, 1.8, 'Iris-virginica'),
       (6.7, 2.5, 5.8, 1.8, 'Iris-virginica'),
       (7.2, 3.6, 6.1, 2.5, 'Iris-virginica'),
       (6.5, 3.2, 5.1, 2. , 'Iris-virginica'),
       (6.4, 2.7, 5.3, 1.9, 'Iris-virginica'),
       (6.8, 3. , 5.5, 2.1, 'Iris-virginica'),
       (5.7, 2.5, 5. , 2. , 'Iris-virginica'),
       (5.8, 2.8, 5.1, 2.4, 'Iris-virginica'),
       (6.4, 3.2, 5.3, 2.3, 'Iris-virginica'),
       (6.5, 3. , 5.5, 1.8, 'Iris-virginica'),
       (7.7, 3.8, 6.7, 2.2, 'Iris-virginica'),
       (7.7, 2.6, 6.9, 2.3, 'Iris-virginica'),
       (6. , 2.2, 5. , 1.5, 'Iris-virginica'),
       (6.9, 3.2, 5.7, 2.3, 'Iris-virginica'),
       (5.6, 2.8, 4.9, 2. , 'Iris-virginica'),
       (7.7, 2.8, 6.7, 2. , 'Iris-virginica'),
       (6.3, 2.7, 4.9, 1.8, 'Iris-virginica'),
       (6.7, 3.3, 5.7, 2.1, 'Iris-virginica'),
       (7.2, 3.2, 6. , 1.8, 'Iris-virginica'),
       (6.2, 2.8, 4.8, 1.8, 'Iris-virginica'),
       (6.1, 3. , 4.9, 1.8, 'Iris-virginica'),
       (6.4, 2.8, 5.6, 2.1, 'Iris-virginica'),
       (7.2, 3. , 5.8, 1.6, 'Iris-virginica'),
       (7.4, 2.8, 6.1, 1.9, 'Iris-virginica'),
       (7.9, 3.8, 6.4, 2. , 'Iris-virginica'),
       (6.4, 2.8, 5.6, 2.2, 'Iris-virginica'),
       (6.3, 2.8, 5.1, 1.5, 'Iris-virginica'),
       (6.1, 2.6, 5.6, 1.4, 'Iris-virginica'),
       (7.7, 3. , 6.1, 2.3, 'Iris-virginica'),
       (6.3, 3.4, 5.6, 2.4, 'Iris-virginica'),
       (6.4, 3.1, 5.5, 1.8, 'Iris-virginica'),
       (6. , 3. , 4.8, 1.8, 'Iris-virginica'),
       (6.9, 3.1, 5.4, 2.1, 'Iris-virginica'),
       (6.7, 3.1, 5.6, 2.4, 'Iris-virginica'),
       (6.9, 3.1, 5.1, 2.3, 'Iris-virginica'),
       (5.8, 2.7, 5.1, 1.9, 'Iris-virginica'),
       (6.8, 3.2, 5.9, 2.3, 'Iris-virginica'),
       (6.7, 3.3, 5.7, 2.5, 'Iris-virginica'),
       (6.7, 3. , 5.2, 2.3, 'Iris-virginica'),
       (6.3, 2.5, 5. , 1.9, 'Iris-virginica'),
       (6.5, 3. , 5.2, 2. , 'Iris-virginica'),
       (6.2, 3.4, 5.4, 2.3, 'Iris-virginica'),
       (5.9, 3. , 5.1, 1.8, 'Iris-virginica')],
      dtype=[('sepal_length', '<f8'), ('sepal_width', '<f8'), ('petal_length', '<f8'), ('petal_width', '<f8'), ('species', '<U15')])
```

In [7…

```
iris_df = pd.DataFrame(iris)
iris_df
```

Out[7…

| | sepal_length | sepal_width | petal_length | petal_width | species |
|---|---|---|---|---|---|
| **0** | 5.1 | 3.5 | 1.4 | 0.2 | Iris-setosa |
| **1** | 4.9 | 3.0 | 1.4 | 0.2 | Iris-setosa |
| **2** | 4.7 | 3.2 | 1.3 | 0.2 | Iris-setosa |
| **3** | 4.6 | 3.1 | 1.5 | 0.2 | Iris-setosa |
| **4** | 5.0 | 3.6 | 1.4 | 0.2 | Iris-setosa |
| **...** | ... | ... | ... | ... | ... |
| **145** | 6.7 | 3.0 | 5.2 | 2.3 | Iris-virginica |
| **146** | 6.3 | 2.5 | 5.0 | 1.9 | Iris-virginica |
| **147** | 6.5 | 3.0 | 5.2 | 2.0 | Iris-virginica |
| **148** | 6.2 | 3.4 | 5.4 | 2.3 | Iris-virginica |
| **149** | 5.9 | 3.0 | 5.1 | 1.8 | Iris-virginica |

150 rows × 5 columns

In [1…

```
iris_df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [1…

```
iris_df.head()
```

Out[1…

| | sepal_length | sepal_width | petal_length | petal_width | species |
|---|---|---|---|---|---|
| **0** | 5.1 | 3.5 | 1.4 | 0.2 | Iris-setosa |
| **1** | 4.9 | 3.0 | 1.4 | 0.2 | Iris-setosa |
| **2** | 4.7 | 3.2 | 1.3 | 0.2 | Iris-setosa |
| **3** | 4.6 | 3.1 | 1.5 | 0.2 | Iris-setosa |
| **4** | 5.0 | 3.6 | 1.4 | 0.2 | Iris-setosa |

In [1…

```
iris_df.tail()
```

Out[1…

| | sepal_length | sepal_width | petal_length | petal_width | species |
|---|---|---|---|---|---|
| **145** | 6.7 | 3.0 | 5.2 | 2.3 | Iris-virginica |
| **146** | 6.3 | 2.5 | 5.0 | 1.9 | Iris-virginica |
| **147** | 6.5 | 3.0 | 5.2 | 2.0 | Iris-virginica |
| **148** | 6.2 | 3.4 | 5.4 | 2.3 | Iris-virginica |
| **149** | 5.9 | 3.0 | 5.1 | 1.8 | Iris-virginica |

In [1…

```
iris_df.describe()
```

Out[1…

| | sepal_length | sepal_width | petal_length | petal_width |
|---|---|---|---|---|
| **count** | 150.000000 | 150.000000 | 150.000000 | 150.000000 |
| **mean** | 5.843333 | 3.054000 | 3.758667 | 1.198667 |
| **std** | 0.828066 | 0.433594 | 1.764420 | 0.763161 |
| **min** | 4.300000 | 2.000000 | 1.000000 | 0.100000 |
| **25%** | 5.100000 | 2.800000 | 1.600000 | 0.300000 |
| **50%** | 5.800000 | 3.000000 | 4.350000 | 1.300000 |
| **75%** | 6.400000 | 3.300000 | 5.100000 | 1.800000 |
| **max** | 7.900000 | 4.400000 | 6.900000 | 2.500000 |

## Select columns

In [1…

```
attrs = ['sepal_length', 'sepal_width']
iris_df[attrs]
```

Out[1…

| | sepal_length | sepal_width |
|---|---|---|
| **0** | 5.1 | 3.5 |
| **1** | 4.9 | 3.0 |
| **2** | 4.7 | 3.2 |
| **3** | 4.6 | 3.1 |
| **4** | 5.0 | 3.6 |
| **...** | ... | ... |
| **145** | 6.7 | 3.0 |
| **146** | 6.3 | 2.5 |
| **147** | 6.5 | 3.0 |
| **148** | 6.2 | 3.4 |
| **149** | 5.9 | 3.0 |

150 rows × 2 columns

In [1…

```
iris_df[attrs].mean()
```

Out[1…

```
sepal_length    5.843333
sepal_width     3.054000
dtype: float64
```

In [1…

```
iris_df[attrs].value_counts()
```

Out[1…

```
sepal_length  sepal_width
5.8           2.7            4
6.5           3.0            3
6.7           3.1            3
6.9           3.1            3
4.9           3.1            3
                            ..
5.5           2.3            1
5.4           3.7            1
              3.0            1
5.3           3.7            1
7.9           3.8            1
Name: count, Length: 116, dtype: int64
```

## Conditions

- Applying operators to columns returns a boolean mask for this column

In [9…

```
iris_df["species"] == "Iris-virginica"
```

Out[9…

```
0      False
1      False
2      False
3      False
4      False
       ...  
145     True
146     True
147     True
148     True
149     True
Name: species, Length: 150, dtype: bool
```

In [9…

```
iris_df["sepal_length"] > 6
```

Out[9…

```
0      False
1      False
2      False
3      False
4      False
       ...  
145     True
146     True
147     True
148     True
149    False
Name: sepal_length, Length: 150, dtype: bool
```

To see only the rows which are TRUE - include the df context around this condition

In [9…

```
iris_df[ iris_df["species"] == "Iris-virginica" ]
```

Out[9…

| | sepal_length | sepal_width | petal_length | petal_width | species |
|---|---|---|---|---|---|
| **100** | 6.3 | 3.3 | 6.0 | 2.5 | Iris-virginica |
| **101** | 5.8 | 2.7 | 5.1 | 1.9 | Iris-virginica |
| **102** | 7.1 | 3.0 | 5.9 | 2.1 | Iris-virginica |
| **103** | 6.3 | 2.9 | 5.6 | 1.8 | Iris-virginica |
| **104** | 6.5 | 3.0 | 5.8 | 2.2 | Iris-virginica |
| **105** | 7.6 | 3.0 | 6.6 | 2.1 | Iris-virginica |
| **106** | 4.9 | 2.5 | 4.5 | 1.7 | Iris-virginica |
| **107** | 7.3 | 2.9 | 6.3 | 1.8 | Iris-virginica |
| **108** | 6.7 | 2.5 | 5.8 | 1.8 | Iris-virginica |
| **109** | 7.2 | 3.6 | 6.1 | 2.5 | Iris-virginica |
| **110** | 6.5 | 3.2 | 5.1 | 2.0 | Iris-virginica |
| **111** | 6.4 | 2.7 | 5.3 | 1.9 | Iris-virginica |
| **112** | 6.8 | 3.0 | 5.5 | 2.1 | Iris-virginica |
| **113** | 5.7 | 2.5 | 5.0 | 2.0 | Iris-virginica |
| **114** | 5.8 | 2.8 | 5.1 | 2.4 | Iris-virginica |
| **115** | 6.4 | 3.2 | 5.3 | 2.3 | Iris-virginica |
| **116** | 6.5 | 3.0 | 5.5 | 1.8 | Iris-virginica |
| **117** | 7.7 | 3.8 | 6.7 | 2.2 | Iris-virginica |
| **118** | 7.7 | 2.6 | 6.9 | 2.3 | Iris-virginica |
| **119** | 6.0 | 2.2 | 5.0 | 1.5 | Iris-virginica |
| **120** | 6.9 | 3.2 | 5.7 | 2.3 | Iris-virginica |
| **121** | 5.6 | 2.8 | 4.9 | 2.0 | Iris-virginica |
| **122** | 7.7 | 2.8 | 6.7 | 2.0 | Iris-virginica |
| **123** | 6.3 | 2.7 | 4.9 | 1.8 | Iris-virginica |
| **124** | 6.7 | 3.3 | 5.7 | 2.1 | Iris-virginica |
| **125** | 7.2 | 3.2 | 6.0 | 1.8 | Iris-virginica |
| **126** | 6.2 | 2.8 | 4.8 | 1.8 | Iris-virginica |
| **127** | 6.1 | 3.0 | 4.9 | 1.8 | Iris-virginica |
| **128** | 6.4 | 2.8 | 5.6 | 2.1 | Iris-virginica |
| **129** | 7.2 | 3.0 | 5.8 | 1.6 | Iris-virginica |
| **130** | 7.4 | 2.8 | 6.1 | 1.9 | Iris-virginica |
| **131** | 7.9 | 3.8 | 6.4 | 2.0 | Iris-virginica |
| **132** | 6.4 | 2.8 | 5.6 | 2.2 | Iris-virginica |
| **133** | 6.3 | 2.8 | 5.1 | 1.5 | Iris-virginica |
| **134** | 6.1 | 2.6 | 5.6 | 1.4 | Iris-virginica |
| **135** | 7.7 | 3.0 | 6.1 | 2.3 | Iris-virginica |
| **136** | 6.3 | 3.4 | 5.6 | 2.4 | Iris-virginica |
| **137** | 6.4 | 3.1 | 5.5 | 1.8 | Iris-virginica |
| **138** | 6.0 | 3.0 | 4.8 | 1.8 | Iris-virginica |
| **139** | 6.9 | 3.1 | 5.4 | 2.1 | Iris-virginica |
| **140** | 6.7 | 3.1 | 5.6 | 2.4 | Iris-virginica |
| **141** | 6.9 | 3.1 | 5.1 | 2.3 | Iris-virginica |
| **142** | 5.8 | 2.7 | 5.1 | 1.9 | Iris-virginica |
| **143** | 6.8 | 3.2 | 5.9 | 2.3 | Iris-virginica |
| **144** | 6.7 | 3.3 | 5.7 | 2.5 | Iris-virginica |
| **145** | 6.7 | 3.0 | 5.2 | 2.3 | Iris-virginica |
| **146** | 6.3 | 2.5 | 5.0 | 1.9 | Iris-virginica |
| **147** | 6.5 | 3.0 | 5.2 | 2.0 | Iris-virginica |
| **148** | 6.2 | 3.4 | 5.4 | 2.3 | Iris-virginica |
| **149** | 5.9 | 3.0 | 5.1 | 1.8 | Iris-virginica |

In [1…

```
iris_df[ iris_df['sepal_length'] > 7 ]
```

Out[1…

| | sepal_length | sepal_width | petal_length | petal_width | species |
|---|---|---|---|---|---|
| **102** | 7.1 | 3.0 | 5.9 | 2.1 | Iris-virginica |
| **105** | 7.6 | 3.0 | 6.6 | 2.1 | Iris-virginica |
| **107** | 7.3 | 2.9 | 6.3 | 1.8 | Iris-virginica |
| **109** | 7.2 | 3.6 | 6.1 | 2.5 | Iris-virginica |
| **117** | 7.7 | 3.8 | 6.7 | 2.2 | Iris-virginica |
| **118** | 7.7 | 2.6 | 6.9 | 2.3 | Iris-virginica |
| **122** | 7.7 | 2.8 | 6.7 | 2.0 | Iris-virginica |
| **125** | 7.2 | 3.2 | 6.0 | 1.8 | Iris-virginica |
| **129** | 7.2 | 3.0 | 5.8 | 1.6 | Iris-virginica |
| **130** | 7.4 | 2.8 | 6.1 | 1.9 | Iris-virginica |
| **131** | 7.9 | 3.8 | 6.4 | 2.0 | Iris-virginica |
| **135** | 7.7 | 3.0 | 6.1 | 2.3 | Iris-virginica |

## How about multiple conditions?

- **Question**: The code below throws an exception... why is this?

In [1…

```
iris_df["species"] == "Iris-virginica" & iris_df['sepal_length'] > 7
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
File ~/Library/Python/3.9/lib/python/site-packages/pandas/core/ops/array_ops.py:362, in na_logical_op(x, y, op)
    353 try:
    354     # For exposition, write:
    355     #  yarr = isinstance(y, np.ndarray)
   (...)
    360     # Then Cases where this goes through without raising include:
    361     #  (xint or xbool) and (yint or bool)
--> 362     result = op(x, y)
    363 except TypeError:

File ~/Library/Python/3.9/lib/python/site-packages/pandas/core/roperator.py:54, in rand_(left, right)
     53 def rand_(left, right):
---> 54     return operator.and_(right, left)

TypeError: ufunc 'bitwise_and' not supported for the input types, and the inputs could not be safely coerced to any supported types acc
ording to the casting rule ''safe''

During handling of the above exception, another exception occurred:

ValueError                                Traceback (most recent call last)
File ~/Library/Python/3.9/lib/python/site-packages/pandas/core/ops/array_ops.py:376, in na_logical_op(x, y, op)
    375 try:
--> 376     result = libops.scalar_binop(x, y, op)
    377 except (
    378     TypeError,
    379     ValueError,
   (...)
    382     NotImplementedError,
    383 ) as err:

File ops.pyx:180, in pandas._libs.ops.scalar_binop()

ValueError: Buffer dtype mismatch, expected 'Python object' but got 'double'

The above exception was the direct cause of the following exception:

TypeError                                 Traceback (most recent call last)
Cell In [102], line 1
----> 1 iris_df["species"] == "Iris-virginica" & iris_df['sepal_length'] > 7

File ~/Library/Python/3.9/lib/python/site-packages/pandas/core/ops/common.py:76, in _unpack_zerodim_and_defer.<locals>.new_method(self,
other)
     72             return NotImplemented
     74 other = item_from_zerodim(other)
---> 76 return method(self, other)

File ~/Library/Python/3.9/lib/python/site-packages/pandas/core/arraylike.py:74, in OpsMixin.__rand__(self, other)
     72 @unpack_zerodim_and_defer("__rand__")
     73 def __rand__(self, other):
---> 74     return self._logical_method(other, roperator.rand_)

File ~/Library/Python/3.9/lib/python/site-packages/pandas/core/series.py:6130, in Series._logical_method(self, other, op)
   6127 lvalues = self._values
   6128 rvalues = extract_array(other, extract_numpy=True, extract_range=True)
-> 6130 res_values = ops.logical_op(lvalues, rvalues, op)
   6131 return self._construct_result(res_values, name=res_name)

File ~/Library/Python/3.9/lib/python/site-packages/pandas/core/ops/array_ops.py:454, in logical_op(left, right, op)
    450 else:
    451     # i.e. scalar
    452     is_other_int_dtype = lib.is_integer(rvalues)
--> 454 res_values = na_logical_op(lvalues, rvalues, op)
    456 # For int vs int `^`, `|`, `&` are bitwise operators and return
    457 #   integer dtypes.  Otherwise these are boolean ops
    458 if not (left.dtype.kind in "iu" and is_other_int_dtype):

File ~/Library/Python/3.9/lib/python/site-packages/pandas/core/ops/array_ops.py:385, in na_logical_op(x, y, op)
    377         except (
    378             TypeError,
    379             ValueError,
   (...)
    382             NotImplementedError,
    383         ) as err:
    384             typ = type(y).__name__
--> 385             raise TypeError(
    386                 f"Cannot perform '{op.__name__}' with a dtyped [{x.dtype}] array "
    387                 f"and scalar of type [{typ}]"
    388             ) from err
    390 return result.reshape(x.shape)

TypeError: Cannot perform 'rand_' with a dtyped [float64] array and scalar of type [bool]
```

In [1…

```
(iris_df["species"] == "Iris-virginica") & (iris_df['sepal_length'] > 7)
```

Out[1…

```
0      False
1      False
2      False
3      False
4      False
       ...  
145    False
146    False
147    False
148    False
149    False
Length: 150, dtype: bool
```

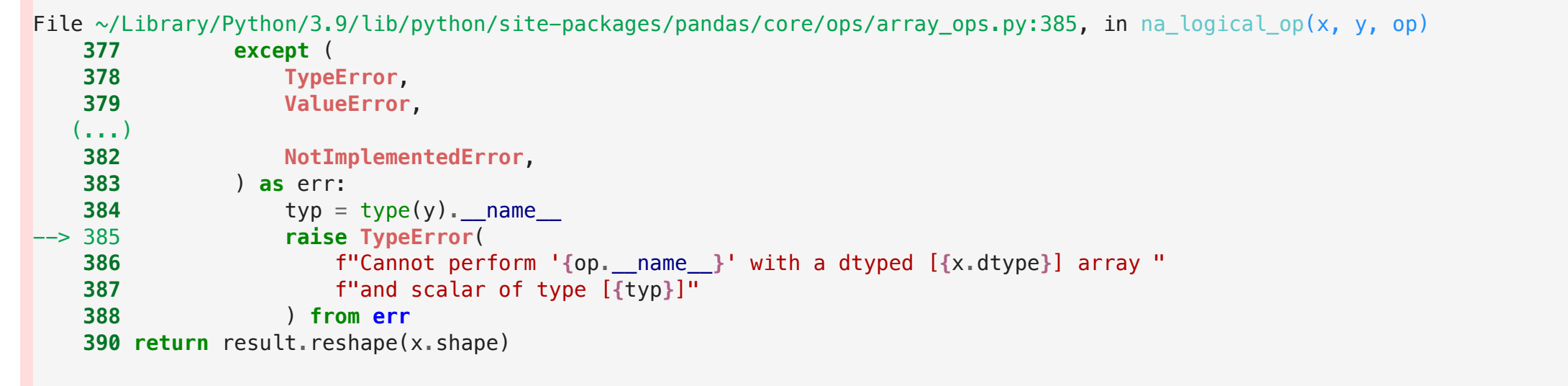
In [1…

```
iris_df[ (iris_df["species"] == "Iris-virginica") & (iris_df['sepal_length'] > 7) ]
```

Out[1…

| | sepal_length | sepal_width | petal_length | petal_width | species |
|---|---|---|---|---|---|
| **102** | 7.1 | 3.0 | 5.9 | 2.1 | Iris-virginica |
| **105** | 7.6 | 3.0 | 6.6 | 2.1 | Iris-virginica |
| **107** | 7.3 | 2.9 | 6.3 | 1.8 | Iris-virginica |
| **109** | 7.2 | 3.6 | 6.1 | 2.5 | Iris-virginica |
| **117** | 7.7 | 3.8 | 6.7 | 2.2 | Iris-virginica |
| **118** | 7.7 | 2.6 | 6.9 | 2.3 | Iris-virginica |
| **122** | 7.7 | 2.8 | 6.7 | 2.0 | Iris-virginica |
| **125** | 7.2 | 3.2 | 6.0 | 1.8 | Iris-virginica |
| **129** | 7.2 | 3.0 | 5.8 | 1.6 | Iris-virginica |
| **130** | 7.4 | 2.8 | 6.1 | 1.9 | Iris-virginica |
| **131** | 7.9 | 3.8 | 6.4 | 2.0 | Iris-virginica |
| **135** | 7.7 | 3.0 | 6.1 | 2.3 | Iris-virginica |

## GroupBy



| | species | sepal_length | sepal_width | petal_length | petal_width |
|---|---|---|---|---|---|
| **0** | setosa | 5.1 | 3.5 | 1.4 | 0.2 |
| **1** | setosa | 4.9 | 3.0 | 1.4 | 0.2 |
| **2** | setosa | 4.7 | 3.2 | 1.3 | 0.2 |
| **3** | setosa | 4.6 | 3.1 | 1.5 | 0.2 |
| **4** | setosa | 5.0 | 3.6 | 1.4 | 0.2 |
| **50** | versicolor | 7.0 | 3.2 | 4.7 | 1.4 |
| **51** | versicolor | 6.4 | 3.2 | 4.5 | 1.5 |
| **52** | versicolor | 6.9 | 3.1 | 4.9 | 1.5 |
| **53** | versicolor | 5.5 | 2.3 | 4.0 | 1.3 |
| **54** | versicolor | 6.5 | 2.8 | 4.6 | 1.5 |
| **100** | virginica | 6.3 | 3.3 | 6.0 | 2.5 |
| **101** | virginica | 5.8 | 2.7 | 5.1 | 1.9 |
| **102** | virginica | 7.1 | 3.0 | 5.9 | 2.1 |
| **103** | virginica | 6.3 | 2.9 | 5.6 | 1.8 |
| **104** | virginica | 6.5 | 3.0 | 5.8 | 2.2 |

| species | sepal_length | sepal_width | petal_length | petal_width |
|---|---|---|---|---|
| **setosa** | 24.3 | 16.4 | 7.0 | 1.0 |
| **versicolor** | 32.3 | 14.6 | 22.7 | 7.2 |
| **virginica** | 32.0 | 14.9 | 28.4 | 10.5 |

In [1…

```
iris_df.groupby('species').sum()
```

Out[1…

| species | sepal_length | sepal_width | petal_length | petal_width |
|---|---|---|---|---|
| **Iris-setosa** | 250.3 | 170.9 | 73.2 | 12.2 |
| **Iris-versicolor** | 296.8 | 138.5 | 213.0 | 66.3 |
| **Iris-virginica** | 329.4 | 148.7 | 277.6 | 101.3 |

In [1…

```
iris_df.groupby('species').mean()
```

Out[1…

| species | sepal_length | sepal_width | petal_length | petal_width |
|---|---|---|---|---|
| **Iris-setosa** | 5.006 | 3.418 | 1.464 | 0.244 |
| **Iris-versicolor** | 5.936 | 2.770 | 4.260 | 1.326 |
| **Iris-virginica** | 6.588 | 2.974 | 5.552 | 2.026 |

In [1…

```
iris_df.groupby('species').min()
```

Out[1…

| species | sepal_length | sepal_width | petal_length | petal_width |
|---|---|---|---|---|
| **Iris-setosa** | 4.3 | 2.3 | 1.0 | 0.1 |
| **Iris-versicolor** | 4.9 | 2.0 | 3.0 | 1.0 |
| **Iris-virginica** | 4.9 | 2.2 | 4.5 | 1.4 |

In [1…

```
iris_df.groupby('species').max()
```

Out[1…

| species | sepal_length | sepal_width | petal_length | petal_width |
|---|---|---|---|---|
| **Iris-setosa** | 5.8 | 4.4 | 1.9 | 0.6 |
| **Iris-versicolor** | 7.0 | 3.4 | 5.1 | 1.8 |
| **Iris-virginica** | 7.9 | 3.8 | 6.9 | 2.5 |

In [1…

```
iris_df.groupby('species').describe()
```

Out[1…

| | sepal_length | | | | | | | | sepal_width | | ... | petal_length | | petal_width | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | count | mean | std | min | 25% | 50% | 75% | max | count | mean | ... | 75% | max | count | mean | std | min | 25% | 50% | 75% | max |
| species | | | | | | | | | | | | | | | | | | | | | |
| **Iris-setosa** | 50.0 | 5.006 | 0.352490 | 4.3 | 4.800 | 5.0 | 5.2 | 5.8 | 50.0 | 3.418 | ... | 1.575 | 1.9 | 50.0 | 0.244 | 0.107210 | 0.1 | 0.2 | 0.2 | 0.3 | 0.6 |
| **Iris-versicolor** | 50.0 | 5.936 | 0.516171 | 4.9 | 5.600 | 5.9 | 6.3 | 7.0 | 50.0 | 2.770 | ... | 4.600 | 5.1 | 50.0 | 1.326 | 0.197753 | 1.0 | 1.2 | 1.3 | 1.5 | 1.8 |
| **Iris-virginica** | 50.0 | 6.588 | 0.635880 | 4.9 | 6.225 | 6.5 | 6.9 | 7.9 | 50.0 | 2.974 | ... | 5.875 | 6.9 | 50.0 | 2.026 | 0.274650 | 1.4 | 1.8 | 2.0 | 2.3 | 2.5 |

3 rows × 32 columns

## EXTRA materials on importing data from APIs - iTunes Example:

Example searches:

- `https://itunes.apple.com/search?term=jack+johnson`
- `https://itunes.apple.com/search?term=jack+johnson&limit=25`
- `https://itunes.apple.com/search?term=queen&limit=25`

Not just music either, iTunes has books, which you can lookup via ISBN:

- `https://itunes.apple.com/lookup?isbn=9780316069359`

For more information and examples of how to construct search queries, see the iTunes documentation here.

Depending on which environment you're running this notebook, you may need to install `requests` before using this package.

```
pip install requests
```

```
python3 -m pip install -U requests --user
```

In [1…

```
import requests
```

In [1…

```
def search_itunes(query, media_type="music"):
    """Search iTunes and return results in JSON format.

    Args:
        query (str): The search term.
        media_type (str): The type of media to search for (e.g., music, movie, podcast).

    Returns:
        dict: The JSON response from the iTunes API.
    """
    base_url = "https://itunes.apple.com/search"
    params = {
        "term": query,
        "media": media_type,
        "limit": 10  # Limit the number of results to 10
    }

    response = requests.get(base_url, params=params)

    # Check if the request was successful
    if response.status_code == 200:
        return response.json()  # Convert the response to JSON
    else:
        response.raise_for_status()  # Raise an exception for HTTP errors
```

In [1…

```
#from search_itunes import search_itunes

df = pd.DataFrame(search_itunes("Python Programming", "ebook"))
```

In [1…

```
df
```

Out[1…

| | resultCount | results |
|---|---|---|
| **0** | 12 | {'fileSizeBytes': 9524154, 'formattedPrice': '... |
| **1** | 12 | {'fileSizeBytes': 3807233, 'formattedPrice': '... |
| **2** | 12 | {'fileSizeBytes': 1693255, 'formattedPrice': '... |
| **3** | 12 | {'fileSizeBytes': 6151269, 'formattedPrice': '... |
| **4** | 12 | {'fileSizeBytes': 305883, 'formattedPrice': '$... |
| **5** | 12 | {'fileSizeBytes': 11014147, 'formattedPrice': ... |
| **6** | 12 | {'fileSizeBytes': 189804, 'formattedPrice': '$... |
| **7** | 12 | {'fileSizeBytes': 154278, 'formattedPrice': '$... |
| **8** | 12 | {'fileSizeBytes': 131835, 'formattedPrice': 'F... |
| **9** | 12 | {'fileSizeBytes': 6164878, 'formattedPrice': '... |
| **10** | 12 | {'fileSizeBytes': 130177, 'formattedPrice': '$... |
| **11** | 12 | {'fileSizeBytes': 5366422, 'formattedPrice': '... |

Due to the data being return as a list of dictionaries.

If we 'zoom in' to the results column, we'll see the `{ }` syntax of the dictionary / json format.

In [1…

```
df['results']
```

Out[1…

```
0    {'currency': 'USD', 'artworkUrl60': 'https://i...
1    {'currency': 'USD', 'artworkUrl60': 'https://i...
2    {'currency': 'USD', 'artworkUrl60': 'https://i...
3    {'currency': 'USD', 'artworkUrl60': 'https://i...
4    {'currency': 'USD', 'artworkUrl60': 'https://i...
5    {'currency': 'USD', 'artworkUrl60': 'https://i...
6    {'currency': 'USD', 'artworkUrl60': 'https://i...
7    {'currency': 'USD', 'artworkUrl60': 'https://i...
8    {'currency': 'USD', 'artworkUrl60': 'https://i...
9    {'currency': 'USD', 'artworkUrl60': 'https://i...
Name: results, dtype: object
```

So how do we extract the values (rows) for each of these keys (which could act as column names) in a 2D matrix...

Well, thankfully there is a handy method named `json_normalise()` which does just the trick!

In [1…

```
df_norm = pd.json_normalize(df['results'])
df_norm
```

Out[1…

| | currency | artworkUrl60 | artworkUrl100 | artistViewUrl | description | tra |
|---|---|---|---|---|---|---|
| **0** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/daniel-corre... | This book is for beginners in programming and ... | 644940 |
| **1** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/ratneshwaran... | Welcome to Python, a programming language whic... | 15242 |
| **2** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/python-softw... | This is an iBooks version of an official Pytho... | 95222 |
| **3** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/bruce-embry/... | Why Python and why another Python book?\t\nSim... | 150008 |
| **4** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/ryan-turner/... | <b>If you're looking for a way to become an ex... | 150824 |
| **5** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/james-tudor/... | Programming Doesn't Have To Be Difficult. If Y... | 155136 |
| **6** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/knowledge-fl... | ★★★★★LEARNING STARTS WITH VIEWING THE WORLD DI... | 110238 |
| **7** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/ryan-turner/... | <b>If you're looking for a complete beginners ... | 150824 |
| **8** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/wikibooks/85... | Now you can easily download your favorite Wiki... | 85727 |
| **9** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/shyam-bharat... | Following topics in Python programming languag... | 142238 |

10 rows × 21 columns

We now have this data in a table, ready for analysis!

In [1…

```
print(df_norm['averageUserRating'].count())
```

```
7
```

In [1…

```
print(df_norm['averageUserRating'].max())
```

```
4.5
```

In [1…

```
print(df_norm['averageUserRating'].mean())
```

```
3.642857142857143
```

In [1…

```
df_norm[df_norm['averageUserRating'] > 3.5]
```

Out[1…

| | currency | artworkUrl60 | artworkUrl100 | artistViewUrl | description | trackId |
|---|---|---|---|---|---|---|
| **0** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/daniel-corre... | This book is for beginners in programming and ... | 6449404731 |
| **2** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/python-softw... | This is an iBooks version of an official Pytho... | 952220899 |
| **3** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/bruce-embry/... | Why Python and why another Python book?\t\nSim... | 1500081797 |
| **8** | USD | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://is1-ssl.mzstatic.com/image/thumb/Publi... | https://books.apple.com/us/artist/wikibooks/85... | Now you can easily download your favorite Wiki... | 857275140 |

4 rows × 21 columns

In [1…

```
df_reduced = df_norm[["trackName", "artistName", "description", "releaseDate", "price", "averageUserRating", "userRatingCount"]]
df_reduced
```

Out[1…

| | trackName | artistName | description | releaseDate | price | averageUserRating | userRatingCount |
|---|---|---|---|---|---|---|---|
| **0** | Python For Beginners: A Practical and Step-by-... | Daniel Correa | This book is for beginners in programming and ... | 2023-05-15T07:00:00Z | 0.00 | 4.5 | 19.0 |
| **1** | Python Simplified | Ratneshwaran Maheswaran | Welcome to Python, a programming language whic... | 2020-07-19T07:00:00Z | 0.00 | 3.5 | 62.0 |
| **2** | Python 3 Tutorial | Python Software Foundation. | This is an iBooks version of an official Pytho... | 2014-12-16T08:00:00Z | 0.00 | 4.0 | 20.0 |
| **3** | Fundamentals of Programming: Using Python | Bruce Embry | Why Python and why another Python book?\t\nSim... | 2020-02-14T08:00:00Z | 0.00 | 4.0 | 7.0 |
| **4** | Python Programming | Ryan Turner | <b>If you're looking for a way to become an ex... | 2020-04-16T07:00:00Z | 9.99 | NaN | NaN |
| **5** | Python Programming For Beginners | James Tudor | Programming Doesn't Have To Be Difficult. If Y... | 2021-01-06T08:00:00Z | 0.00 | 2.0 | 6.0 |
| **6** | Python Programming | Knowledge flow | ★★★★★LEARNING STARTS WITH VIEWING THE WORLD DI... | 2016-04-11T07:00:00Z | 0.99 | NaN | NaN |
| **7** | Python Programming | Ryan Turner | <b>If you're looking for a complete beginners ... | 2020-04-16T07:00:00Z | 2.99 | NaN | NaN |
| **8** | Non-Programmer's Tutorial for Python 3 | Wikibooks | Now you can easily download your favorite Wiki... | 2014-04-06T21:47:50Z | 0.00 | 4.0 | 31.0 |
| **9** | Learn Python | Shyam Bharath, S.D. | Following topics in Python programming languag... | 2018-08-04T07:00:00Z | 0.00 | 3.5 | 24.0 |

In [1…

```
df_reduced[df_reduced['averageUserRating'] > 3.5]
```

Out[1…

| | trackName | artistName | description | releaseDate | price | averageUserRating | userRatingCount |
|---|---|---|---|---|---|---|---|
| **0** | Python For Beginners: A Practical and Step-by-... | Daniel Correa | This book is for beginners in programming and ... | 2023-05-15T07:00:00Z | 0.0 | 4.5 | 19.0 |
| **2** | Python 3 Tutorial | Python Software Foundation. | This is an iBooks version of an official Pytho... | 2014-12-16T08:00:00Z | 0.0 | 4.0 | 20.0 |
| **3** | Fundamentals of Programming: Using Python | Bruce Embry | Why Python and why another Python book?\t\nSim... | 2020-02-14T08:00:00Z | 0.0 | 4.0 | 7.0 |
| **8** | Non-Programmer's Tutorial for Python 3 | Wikibooks | Now you can easily download your favorite Wiki... | 2014-04-06T21:47:50Z | 0.0 | 4.0 | 31.0 |

### This Jupyter Notebook contains exercises for you to extend your introduction to the basics with Python Pandas. Attempt the following exercises, which slowly build in complexity. If you get stuck, check back to the Python lecture recording on Pandas here or view the W3Schools page on Python Pandas, which includes examples, exercises and quizzes to help your understanding.

`pip install numpy`
`pip install pandas`
`pip install matplotlib`
`pip install seaborn`

# 02 Pandas Ecommerce Purchases Exercise (by Jose Portila)

In this Exercise you will be given some Fake Data about some purchases done through Amazon! Just go ahead and follow the directions and try your best to answer the questions and complete the tasks. Feel free to reference the solutions. Most of the tasks can be solved in different ways. For the most part, the questions get progressively harder.

Please excuse anything that doesn't make "Real-World" sense in the dataframe, all the data is fake and made-up.

Also note that all of these questions can be answered with one line of code.

** Import pandas and read in the Ecommerce Purchases csv file and set it to a DataFrame called ecom. **

First check that you have access to `pandas`, by running the cell below:

In [ …

```
#import pandas as pd
```

Out[ …

```
<module 'pandas' from '/Users/nick/Library/Python/3.9/lib/python/site-packages/pandas/__init__.py'>
```

Assuming you downloaded the accompanying 'Ecommerce Purchases.csv' file, check that you can read this in as a Pandas DataFrame.

In [ …

```
... = pd.read_csv("Ecommerce Purchases.csv")
#type(df)
```

```
pandas.core.frame.DataFrame
```

## Exercise 1

**Check the head of the DataFrame.**

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

| | Address | Lot | AM or PM | Browser Info | Company | Credit Card | CC Exp Date | CC Security Code | CC Provider | Email | Job | IP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 16629 Pace Camp Apt. 448\nAlexisborough, NE 77... | 46 in | PM | Opera/9.56. (X11; Linux x86_64; sl-SI) Presto/2... | Martinez-Herman | 6011929061123406 | 02/20 | 900 | JCB 16 digit | pdunlap@yahoo.com | Scientist, product/process development | 149.14 |
| **1** | 9374 Jasmine Spurs Suite 508\nSouth John, TN 8... | 28 rn | PM | Opera/8.93. (Windows 98; Win 9x 4.90; en-US) Pr... | Fletcher, Richards and Whitaker | 3337758169645356 | 11/18 | 561 | Mastercard | anthony41@reed.com | Drilling engineer | 15. |
| **2** | Unit 0065 Box 5052\nDPO AP 27450 | 94 vE | PM | Mozilla/5.0 (compatible; MSIE 9.0; Windows NT ... | Simpson, Williams and Pham | 675957666125 | 08/19 | 699 | JCB 16 digit | amymiller@morales-harrison.com | Customer service manager | 132.2 |
| **3** | 7780 Julia Fords\nNew Stacy, WA 45798 | 36 vm | PM | Mozilla/5.0 (Macintosh; Intel Mac OS X 10_8_0 ... | Williams, Marshall and Buchanan | 6011578504430710 | 02/24 | 384 | Discover | brent16@olson-robinson.info | Drilling engineer | 30. |
| **4** | 23012 Munoz Drive Suite 337\nNew Cynthia, TX 5... | 20 IE | AM | Opera/9.58. (X11; Linux x86_64; it-IT) Presto/2... | Brown, Watson and Andrews | 6011456623207998 | 10/25 | 678 | Diners Club / Carte Blanche | christopherwright@gmail.com | Fine artist | 24. |

## Exercise 2

** How many rows and columns are there? **

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 14 columns):
Address             10000 non-null object
Lot                 10000 non-null object
AM or PM            10000 non-null object
Browser Info        10000 non-null object
Company             10000 non-null object
Credit Card         10000 non-null int64
CC Exp Date         10000 non-null object
CC Security Code    10000 non-null int64
CC Provider         10000 non-null object
Email               10000 non-null object
Job                 10000 non-null object
IP Address          10000 non-null object
Language            10000 non-null object
Purchase Price      10000 non-null float64
dtypes: float64(1), int64(2), object(11)
memory usage: 1.1+ MB
```

## Exercise 3

** What is the average Purchase Price? **

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
50.34730200000025
```

## Exercise 4

** What were the highest and lowest purchase prices? **

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
99.98999999999995
```

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
0.0
```

## Exercise 5

** How many people have English 'en' as their Language of choice on the website? **

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
1098
```

## Exercise 6

** How many people have the job title of "Lawyer" ? **

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
np.int64(30)
```

## Exercise 7

** How many people made the purchase during the AM and how many people made the purchase during PM ? **

**(Hint: Check out value_counts() ) **

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
PM    5068
AM    4932
Name: AM or PM, dtype: int64
```

## Exercise 8

** What are the 5 most common Job Titles? **

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
Interior and spatial designer    31
Lawyer                           30
Social researcher                28
Purchasing manager               27
Designer, jewellery              27
Name: Job, dtype: int64
```

## Exercise 9

** Someone made a purchase that came from Lot: "90 WT" , what was the Purchase Price for this transaction? **

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
513    75.1
Name: Purchase Price, dtype: float64
```

## Exercise 10

** What is the email of the person with the following Credit Card Number: 4926535242672853 **

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
1234    bondellen@williams-garza.com
Name: Email, dtype: object
```

## Exercise 11

** How many people have American Express as their Credit Card Provider *and* made a purchase above $95 ?**

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
Address             39
Lot                 39
AM or PM            39
Browser Info        39
Company             39
Credit Card         39
CC Exp Date         39
CC Security Code    39
CC Provider         39
Email               39
Job                 39
IP Address          39
Language            39
Purchase Price      39
dtype: int64
```

## Exercise 12

** Hard: How many people have a credit card that expires in 2025? **

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
1033
```

## Exercise 13

** Hard: What are the top 5 most popular email providers/hosts (e.g. gmail.com, yahoo.com, etc...) **

In [ …

```
# Create a new 'code cell' above
# Call a Pandas method to create the view you see below:
```

```
hotmail.com     1638
yahoo.com       1616
gmail.com       1605
smith.com         42
williams.com      37
Name: Email, dtype: int64
```

In [ …

# ADDITIONAL EXERCISES

## Exercise 14

Read in the iris dataset as a numpy array with the code below. Can you extract the three Iris flower labels from the 'species' column ('Iris-setosa', 'Iris-veriscolor', and 'Iris-virginica') and make these keys in a dictionary, which map to a `list` of `tuples` representing each row?

In [4…

```
url = 'https://archive.ics.uci.edu/ml/machine-learning-databases/iris/iris.data'
iris = np.genfromtxt(url, delimiter=',',
names=['sepal length', 'sepal width', 'petal length' , 'petal width', 'species'],
dtype=[np. float64, np.float64, np.float64, np.float64, 'U15']
)
iris
```

Out[4…

```
array([(5.1, 3.5, 1.4, 0.2, 'Iris-setosa'),
       (4.9, 3. , 1.4, 0.2, 'Iris-setosa'),
       (4.7, 3.2, 1.3, 0.2, 'Iris-setosa'),
       (4.6, 3.1, 1.5, 0.2, 'Iris-setosa'),
       (5. , 3.6, 1.4, 0.2, 'Iris-setosa'),
       (5.4, 3.9, 1.7, 0.4, 'Iris-setosa'),
       (4.6, 3.4, 1.4, 0.3, 'Iris-setosa'),
       (5. , 3.4, 1.5, 0.2, 'Iris-setosa'),
       (4.4, 2.9, 1.4, 0.2, 'Iris-setosa'),
       (4.9, 3.1, 1.5, 0.1, 'Iris-setosa'),
       (5.4, 3.7, 1.5, 0.2, 'Iris-setosa'),
       (4.8, 3.4, 1.6, 0.2, 'Iris-setosa'),
       (4.8, 3. , 1.4, 0.1, 'Iris-setosa'),
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       (5.8, 4. , 1.2, 0.2, 'Iris-setosa'),
       (5.7, 4.4, 1.5, 0.4, 'Iris-setosa'),
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       (5.2, 3.4, 1.4, 0.2, 'Iris-setosa'),
       (4.7, 3.2, 1.6, 0.2, 'Iris-setosa'),
       (4.8, 3.1, 1.6, 0.2, 'Iris-setosa'),
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       (4.8, 3. , 1.4, 0.3, 'Iris-setosa'),
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       (7. , 3.2, 4.7, 1.4, 'Iris-versicolor'),
       (6.4, 3.2, 4.5, 1.5, 'Iris-versicolor'),
       (6.9, 3.1, 4.9, 1.5, 'Iris-versicolor'),
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       (5.7, 2.8, 4.5, 1.3, 'Iris-versicolor'),
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       (4.9, 2.4, 3.3, 1. , 'Iris-versicolor'),
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       (5.2, 2.7, 3.9, 1.4, 'Iris-versicolor'),
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       (6. , 2.2, 4. , 1. , 'Iris-versicolor'),
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       (6.7, 3.1, 4.4, 1.4, 'Iris-versicolor'),
       (5.6, 3. , 4.5, 1.5, 'Iris-versicolor'),
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       (6.1, 2.8, 4. , 1.3, 'Iris-versicolor'),
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       (6.1, 2.8, 4.7, 1.2, 'Iris-versicolor'),
       (6.4, 2.9, 4.3, 1.3, 'Iris-versicolor'),
       (6.6, 3. , 4.4, 1.4, 'Iris-versicolor'),
       (6.8, 2.8, 4.8, 1.4, 'Iris-versicolor'),
       (6.7, 3. , 5. , 1.7, 'Iris-versicolor'),
       (6. , 2.9, 4.5, 1.5, 'Iris-versicolor'),
       (5.7, 2.6, 3.5, 1. , 'Iris-versicolor'),
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       (6. , 3.4, 4.5, 1.6, 'Iris-versicolor'),
       (6.7, 3.1, 4.7, 1.5, 'Iris-versicolor'),
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       (5.8, 2.6, 4. , 1.2, 'Iris-versicolor'),
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       (5.7, 2.8, 4.1, 1.3, 'Iris-versicolor'),
       (6.3, 3.3, 6. , 2.5, 'Iris-virginica'),
       (5.8, 2.7, 5.1, 1.9, 'Iris-virginica'),
       (7.1, 3. , 5.9, 2.1, 'Iris-virginica'),
       (6.3, 2.9, 5.6, 1.8, 'Iris-virginica'),
       (6.5, 3. , 5.8, 2.2, 'Iris-virginica'),
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       (6.4, 2.7, 5.3, 1.9, 'Iris-virginica'),
       (6.8, 3. , 5.5, 2.1, 'Iris-virginica'),
       (5.7, 2.5, 5. , 2. , 'Iris-virginica'),
       (5.8, 2.8, 5.1, 2.4, 'Iris-virginica'),
       (6.4, 3.2, 5.3, 2.3, 'Iris-virginica'),
       (6.5, 3. , 5.5, 1.8, 'Iris-virginica'),
       (7.7, 3.8, 6.7, 2.2, 'Iris-virginica'),
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       (7.7, 2.8, 6.7, 2. , 'Iris-virginica'),
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       (6.7, 3.3, 5.7, 2.1, 'Iris-virginica'),
       (7.2, 3.2, 6. , 1.8, 'Iris-virginica'),
       (6.2, 2.8, 4.8, 1.8, 'Iris-virginica'),
       (6.1, 3. , 4.9, 1.8, 'Iris-virginica'),
       (6.4, 2.8, 5.6, 2.1, 'Iris-virginica'),
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       (7.4, 2.8, 6.1, 1.9, 'Iris-virginica'),
       (7.9, 3.8, 6.4, 2. , 'Iris-virginica'),
       (6.4, 2.8, 5.6, 2.2, 'Iris-virginica'),
       (6.3, 2.8, 5.1, 1.5, 'Iris-virginica'),
       (6.1, 2.6, 5.6, 1.4, 'Iris-virginica'),
       (7.7, 3. , 6.1, 2.3, 'Iris-virginica'),
       (6.3, 3.4, 5.6, 2.4, 'Iris-virginica'),
       (6.4, 3.1, 5.5, 1.8, 'Iris-virginica'),
       (6. , 3. , 4.8, 1.8, 'Iris-virginica'),
       (6.9, 3.1, 5.4, 2.1, 'Iris-virginica'),
       (6.7, 3.1, 5.6, 2.4, 'Iris-virginica'),
       (6.9, 3.1, 5.1, 2.3, 'Iris-virginica'),
       (5.8, 2.7, 5.1, 1.9, 'Iris-virginica'),
       (6.8, 3.2, 5.9, 2.3, 'Iris-virginica'),
       (6.7, 3.3, 5.7, 2.5, 'Iris-virginica'),
       (6.7, 3. , 5.2, 2.3, 'Iris-virginica'),
       (6.3, 2.5, 5. , 1.9, 'Iris-virginica'),
       (6.5, 3. , 5.2, 2. , 'Iris-virginica'),
       (6.2, 3.4, 5.4, 2.3, 'Iris-virginica'),
       (5.9, 3. , 5.1, 1.8, 'Iris-virginica')],
      dtype=[('sepal_length', '<f8'), ('sepal_width', '<f8'), ('petal_length', '<f8'), ('petal_width', '<f8'), ('species', '<U15')])
```

Below is an example of how the keys and values should be arranged:
See if you can find an efficient way to organise an iris dictionary in this way.

In [6…

```
# Write your solution here.
```

Out[6…

```
{'Iris-setosa': [(5.1, 3.5, 1.4, 0.2), (4.9, 3.0, 1.4, 0.2), Ellipsis],
 'Iris-veriscolor': [(7.0, 3.2, 4.7, 1.4), (6.4, 3.2, 4.5, 1.5), Ellipsis],
 'Iris-virginica': [(6.3, 3.3, 6.0, 2.5), (5.8, 2.7, 5.1, 1.9), Ellipsis]}
```

Extension: With the dictionary formatted this way, now produce some descriptive statistics for each of the three flowers - minimum and maximum values, and the mean values.

In [ …

```
# Write your solution here.
```

## Exercise 15

Revisit the `search_itunes()` code from the `Extra` section in this notebook.

In [2…

```
import requests
```

```
/Users/nick/Library/Python/3.9/lib/python/site-packages/urllib3/__init__.py:35: NotOpenSSLWarning: urllib3 v2 only supports OpenSSL 1.
1.1+, currently the 'ssl' module is compiled with 'LibreSSL 2.8.3'. See: https://github.com/urllib3/urllib3/issues/3020
  warnings.warn(
```

In [2…

```
def search_itunes(query, media_type="music"):
    """Search iTunes and return results in JSON format.

    Args:
        query (str): The search term.
        media_type (str): The type of media to search for (e.g., music, movie, podcast).

    Returns:
        dict: The JSON response from the iTunes API.
    """
    base_url = "https://itunes.apple.com/search"
    params = {
        "term": query,
        "media": media_type,
        "limit": 10  # Limit the number of results to 10
    }

    response = requests.get(base_url, params=params)

    # Check if the request was successful
    if response.status_code == 200:
        return response.json()  # Convert the response to JSON
    else:
        response.raise_for_status()  # Raise an exception for HTTP errors
```

In [2…

```
query = "Taylor Swift"
media_type = "music"
results = search_itunes(query, media_type)

results
```

Out[2…

```
{'resultCount': 12,
 'results': [{'wrapperType': 'track',
   'kind': 'song',
   'artistId': 159260351,
   'collectionId': 1468058165,
   'trackId': 1468058173,
   'artistName': 'Taylor Swift',
   'collectionName': 'Lover',
   'trackName': 'Lover',
   'collectionCensoredName': 'Lover',
   'trackCensoredName': 'Lover',
   'artistViewUrl': 'https://music.apple.com/us/artist/taylor-swift/159260351?uo=4',
   'collectionViewUrl': 'https://music.apple.com/us/album/lover/1468058165?i=1468058173&uo=4',
   'trackViewUrl': 'https://music.apple.com/us/album/lover/1468058165?i=1468058173&uo=4',
   'previewUrl': 'https://audio-ssl.itunes.apple.com/itunes-assets/AudioPreview211/v4/4f/fd/c7/4ffdc746-c0de-999b-eb93-2753eaa18978/mza
f_8574966813156057641.plus.aac.p.m4a',
   'artworkUrl30': 'https://is1-ssl.mzstatic.com/image/thumb/Music125/v4/49/3d/ab/493dab54-f920-9043-6181-80993b8116c9/19UMGIM53909.rg
b.jpg/30x30bb.jpg',
   'artworkUrl60': 'https://is1-ssl.mzstatic.com/image/thumb/Music125/v4/49/3d/ab/493dab54-f920-9043-6181-80993b8116c9/19UMGIM53909.rg
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b.jpg/100x100bb.jpg',
   'collectionPrice': 11.99,
   'trackPrice': 1.29,
   'releaseDate': '2019-08-16T07:00:00Z',
   'collectionExplicitness': 'notExplicit',
   'trackExplicitness': 'notExplicit',
   'discCount': 1,
   'discNumber': 1,
   'trackCount': 18,
   'trackNumber': 3,
   'trackTimeMillis': 221300,
   'country': 'USA',
   'currency': 'USD',
   'primaryGenreName': 'Pop',
   'isStreamable': True},
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   'kind': 'song',
   'artistId': 159260351,
   'collectionId': 1468058165,
   'trackId': 1468058704,
   'artistName': 'Taylor Swift',
   'collectionName': 'Lover',
   'trackName': 'You Need To Calm Down',
   'collectionCensoredName': 'Lover',
   'trackCensoredName': 'You Need To Calm Down',
   'artistViewUrl': 'https://music.apple.com/us/artist/taylor-swift/159260351?uo=4',
   'collectionViewUrl': 'https://music.apple.com/us/album/you-need-to-calm-down/1468058165?i=1468058704&uo=4',
   'trackViewUrl': 'https://music.apple.com/us/album/you-need-to-calm-down/1468058165?i=1468058704&uo=4',
   'previewUrl': 'https://audio-ssl.itunes.apple.com/itunes-assets/AudioPreview211/v4/85/05/a9/8505a97c-15ff-f665-fec4-f6bf59e5b0fa/mza
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   'discNumber': 1,
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   'trackNumber': 14,
   'trackTimeMillis': 171352,
   'country': 'USA',
   'currency': 'USD',
   'primaryGenreName': 'Pop',
   'isStreamable': True},
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   'kind': 'song',
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   'collectionId': 1468058165,
   'trackId': 1468058171,
   'artistName': 'Taylor Swift',
   'collectionName': 'Lover',
   'trackName': 'Cruel Summer',
   'collectionCensoredName': 'Lover',
   'trackCensoredName': 'Cruel Summer',
   'artistViewUrl': 'https://music.apple.com/us/artist/taylor-swift/159260351?uo=4',
   'collectionViewUrl': 'https://music.apple.com/us/album/cruel-summer/1468058165?i=1468058171&uo=4',
   'trackViewUrl': 'https://music.apple.com/us/album/cruel-summer/1468058165?i=1468058171&uo=4',
   'previewUrl': 'https://audio-ssl.itunes.apple.com/itunes-assets/AudioPreview211/v4/96/61/ff/9661ff73-0d57-582a-5e57-5687abb33313/mza
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   'trackExplicitness': 'notExplicit',
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   'discNumber': 1,
   'trackCount': 18,
   'trackNumber': 2,
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   'currency': 'USD',
   'primaryGenreName': 'Pop',
   'isStreamable': True},
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   'collectionId': 1468058165,
   'trackId': 1468058706,
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   'collectionName': 'Lover',
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   'trackCensoredName': 'ME! (feat. Brendon Urie of Panic! At The Disco)',
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   'collectionViewUrl': 'https://music.apple.com/us/album/me-feat-brendon-urie-of-panic-at-the-disco/1468058165?i=1468058706&uo=4',
   'trackViewUrl': 'https://music.apple.com/us/album/me-feat-brendon-urie-of-panic-at-the-disco/1468058165?i=1468058706&uo=4',
   'previewUrl': 'https://audio-ssl.itunes.apple.com/itunes-assets/AudioPreview221/v4/08/b2/a1/08b2a176-be4a-6218-3339-dd1dbf00c443/mza
f_13396297911439214793.plus.aac.p.m4a',
   'artworkUrl30': 'https://is1-ssl.mzstatic.com/image/thumb/Music125/v4/49/3d/ab/493dab54-f920-9043-6181-80993b8116c9/19UMGIM53909.rg
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b.jpg/60x60bb.jpg',
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   'currency': 'USD',
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   'isStreamable': True},
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   'trackViewUrl': 'https://music.apple.com/us/album/daylight/1468058165?i=1468058708&uo=4',
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   'trackCount': 18,
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   'currency': 'USD',
   'primaryGenreName': 'Pop',
   'isStreamable': True}]}
```

See below the attempt to format the data returned from the `search_itunes()` function to format the data into a DataFrame.

Question: What do you notice about the results column below?

In [3…

```python
#from search_itunes import search_itunes #only if you have a .py script for search_itunes

df = pd.DataFrame(search_itunes("Taylor", "music"))
df
```

Out[3…

| | resultCount | results |
|---|---|---|
| **0** | 10 | {'wrapperType': 'track', 'kind': 'song', 'arti... |
| **1** | 10 | {'wrapperType': 'track', 'kind': 'song', 'arti... |
| **2** | 10 | {'wrapperType': 'track', 'kind': 'song', 'arti... |
| **3** | 10 | {'wrapperType': 'track', 'kind': 'song', 'arti... |
| **4** | 10 | {'wrapperType': 'track', 'kind': 'song', 'arti... |
| **5** | 10 | {'wrapperType': 'track', 'kind': 'song', 'arti... |
| **6** | 10 | {'wrapperType': 'track', 'kind': 'song', 'arti... |
| **7** | 10 | {'wrapperType': 'track', 'kind': 'song', 'arti... |
| **8** | 10 | {'wrapperType': 'track', 'kind': 'song', 'arti... |
| **9** | 10 | {'wrapperType': 'track', 'kind': 'song', 'arti... |

Your task is to transform the above DataFrame into the format shown below:

In [3…

```python
# Write your solution here...
```

Out[3…

| | wrapperType | kind | artistId | collectionId | trackId | artistName | collectionName | trackName | collectionCensoredName | trackCensoredName | ... | di |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | track | song | 400555769 | 1506831755 | 1506832357 | Natalie Taylor | Surrender - Single | Surrender | Surrender - Single | Surrender | ... | |
| **1** | track | song | 400555769 | 1182561824 | 1182561980 | Natalie Taylor | In the Air Tonight - Single | In the Air Tonight | In the Air Tonight - Single | In the Air Tonight | ... | |
| **2** | track | song | 400555769 | 880114237 | 880114240 | Natalie Taylor | Love Life - Single | Love Life | Love Life - Single | Love Life | ... | |
| **3** | track | song | 400555769 | 1519882762 | 1519882765 | Natalie Taylor | Surrender (Kina Remix) - Single | Surrender (Kina Remix) | Surrender (Kina Remix) - Single | Surrender (Kina Remix) | ... | |
| **4** | track | song | 400555769 | 1178290096 | 1178290111 | Natalie Taylor | Loved You Before - Single | Loved You Before | Loved You Before - Single | Loved You Before | ... | |
| **5** | track | song | 645420096 | 1572859144 | 1572859302 | ILLENIUM & Natalie Taylor | Fallen Embers | Fragments | Fallen Embers | Fragments | ... | |
| **6** | track | song | 400555769 | 978021284 | 978021290 | Natalie Taylor | Wildfire - EP | Wildfire | Wildfire - EP | Wildfire | ... | |
| **7** | track | song | 400555769 | 1457378386 | 1457378387 | Natalie Taylor | Iris - Single | Iris | Iris - Single | Iris | ... | |
| **8** | track | song | 400555769 | 1154603331 | 1154603480 | Natalie Taylor | Amen - Single | Amen | Amen - Single | Amen | ... | |
| **9** | track | song | 400555769 | 988419300 | 988419846 | Natalie Taylor | I Want It All - Single | I Want It All | I Want It All - Single | I Want It All | ... | |

10 rows × 32 columns

Extension: Write code which returns the song with the longest `trackTime`

Hint: Start by identifying the column that has data on the `trackTime`

In [ …

```python
```

## Exercise 16

Find an CSV file online, that you can import via `read_csv()`. Be aware that the data needs to be in raw format (without the html/css/table structure). Alternatively, download the CSV file and read it locally (as some sites won't give you the raw data in a readable/scrapable format).

Examples of CSV data:

https://github.com/riteshc6/amazon_scraper/blob/master/downloads/Computers%20%26%20Accessories.csv

https://raw.githubusercontent.com/riteshc6/amazon_scraper/master/downloads/Computers%20%26%20Accessories.csv

In [1…

```python
url = "https://raw.githubusercontent.com/riteshc6/amazon_scraper/master/downloads/Computers%20%26%20Accessories.csv"

df = pd.read_csv(url)

df
```

Inspect the data from your chosen CSV, and perform some exploratory data analysis.
Question: What trends do you see?

This website does not host notebooks, it only renders notebooks available on other websites.

Delivered by Fastly, Rendered by OVHcloud

nbviewer GitHub repository.

nbconvert version: 7.16.6

Rendered a few seconds ago